

Lec 5

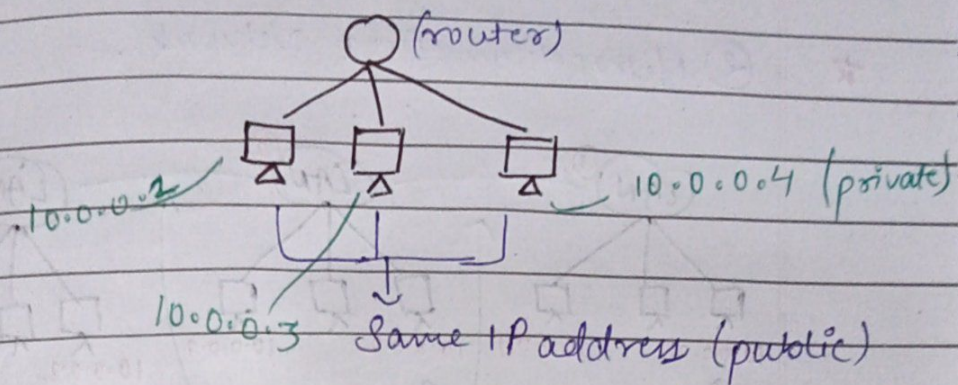
Basic of Internet

Topology + VPN

Date 20/9/24
Page

Types of Network

① Local Area Network



— in phone, laptop we will always see public IP address.

— private address router deta hai.

ye prt. add.
isliye bnaya kuki IPv4
limited h. (400 cr)

→ if bahar se data aata hai to us network me router ko pata hoga ki kisme request kii thi → to data us particular device k paas jayegi. (private 'ip address')

NAT → "Network Address Translator"

How data travels all over world?

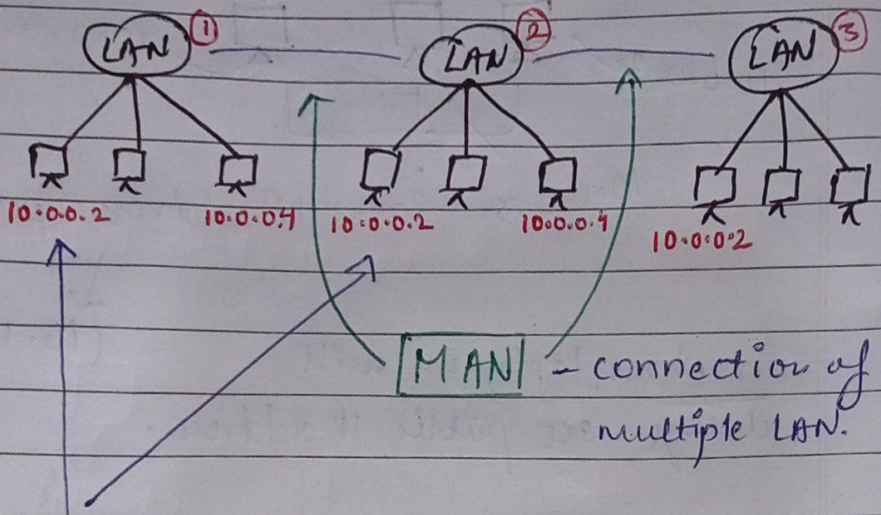
— under sea wires (fastest - country to country)

private IP address

- ↳ Class A } $10.0.0.0 \longleftrightarrow 10.255.255.255$
- ↳ Class B } $172.16.0.0 \longleftrightarrow 172.31.255.255$
- ↳ Class C } $192.168.0.0 \longleftrightarrow 192.168.255.255$

★

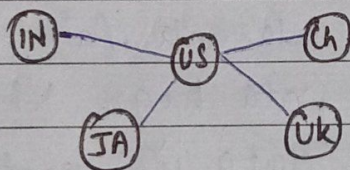
② Metropolitan Area Network



— here private ip can be same within one & second lan.

★

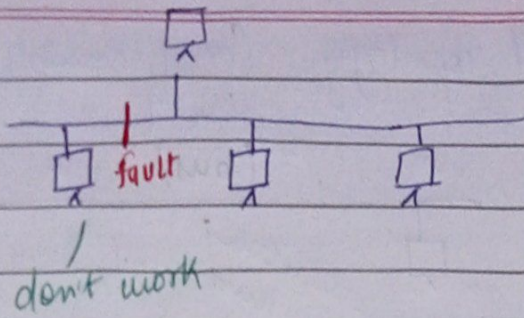
③ Wide Area Network



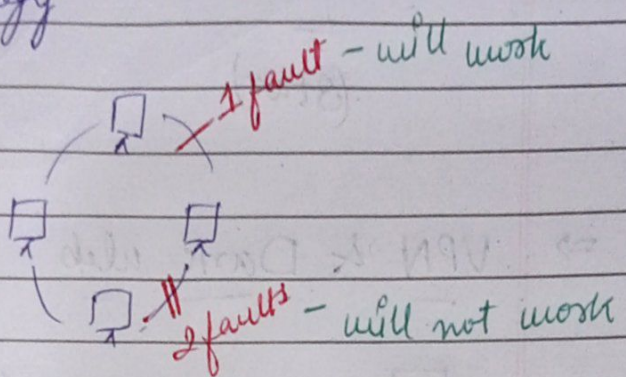
country to country

= Types of Topologies

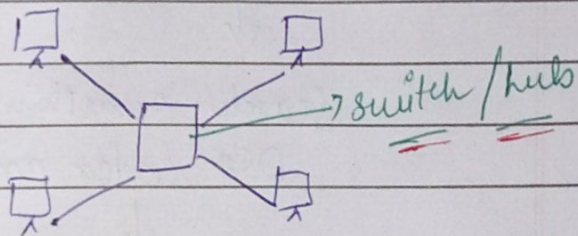
① Bus Topology.



(2) Ring Topology

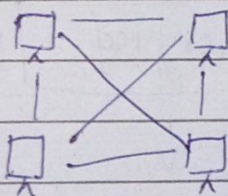


(3) star topology



- data will be transmitted through switch/hubs.
- switch damage $\rightarrow$ everything stops.!

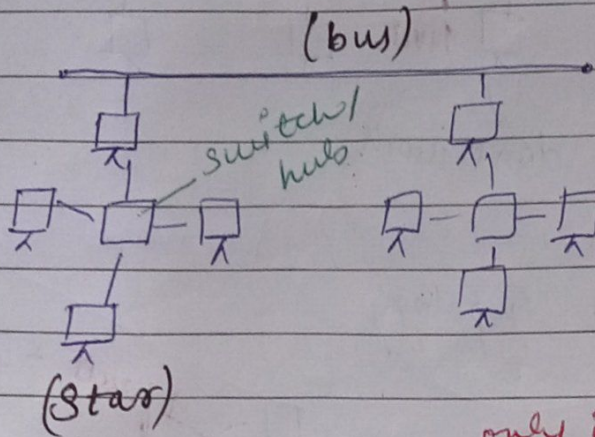
(4) Mesh Topology



no problem if
fault there, are
multiple routes.

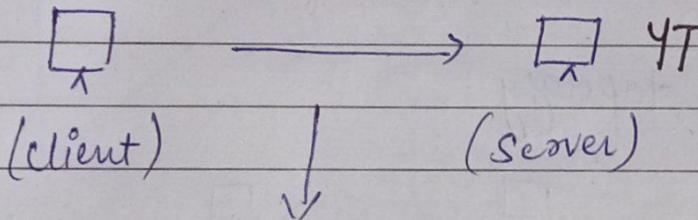
- problem while adding new system

(5) Hybrid Topology (any combination)

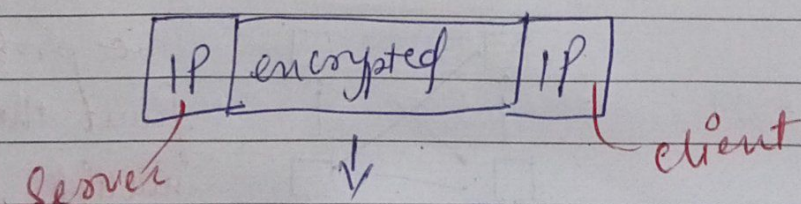
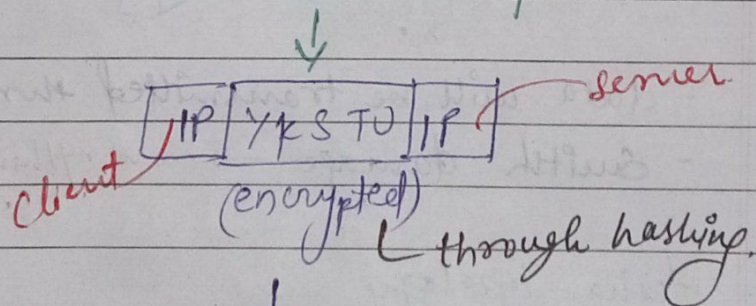


only in LAN

⇒ VPN & Dark web



Search something
DSA coder army



DSA Coder army
comes to original text

→ ISP knows you visited YT but (search not) ^{doesn't know}

Inognito Mode

ban but only domain name (easiest method)

↳ everything can be tracked by ~~ISP~~ ISP.

↳ search history / browse history is not saved on phone.

पता चापता पता

VPN (virtual private network)

→ to use websites not available in India (or banned)

 ~~21.6.6.5~~ → india ^{changed to US-IP address}

(me) 76.28.5.9 → USA

ISP doesn't allow to go to ban website.

ban → 12.6.6.9

xyz.com

IP address

Me	xyz.com	VPN
----	---------	-----

 IP address

encrypt

so ISP allows to access by seeing VPN request for xyz.com.

Then VPN goes instead of us to the ban website.

